

Formalised Aims and Objectives

Aim: The project aims to identify the most effective machine learning model for plant identification in real-time using smart glasses.

Objectives:

- To gather a dataset of at least 5 different plant species, by the end of October.
- To train and evaluate at least 3 different models for plant identification accuracy.
- To optimize the models for real-time use through the specified smart glasses
- To have a specific smart glass made and optimized for connection to an external device, for processing in real-time.
- To complete a comparative analysis of the model's accuracy.

Methodology

This project will follow a quantitative methodology to comparatively analyse machine learning models. The objective is to train, optimise and test a range of machine learning model to evaluate their capability to identify plant species in real-time. With them being tested and simulated on a resource-constrained hardware which will be smart glasses.

The methodology will be structured as follows:

1. Dataset Selection and Preparation

A publicly available dataset containing at least five distinct plant species will be selected. The Images will then be pre-processed to ensure consistency. This will include; Resizing to a standard dimension, Normalization of pixels, data augmentation to improve model generalisation. These dataset images will then be divided into training, validation and testing subsets to enable a fair evaluation.

2. Model Selection and Training

At least three separate machine learning models will be selected for comparison. Each model will be trained using the specific dataset selected earlier to ensure fair comparisons are made. Which may include; A convolutional neural network model, a pre-trained learning model (e.g. MobileNet, EfficientNet), a lightweight model optimized for edge devices.

3. Model Evaluation

The models will then be evaluated to determine their suitability for real-time use. Done using the following metrics:

Overall accuracy, Precision, Recall, Inference speed, Model size

4. Optimisation and integration for real-time use involving the smart glasses

The best performing models or all models will then be optimised for use on the smart glasses. This will involve converting the models to a format that is suitable for edge devices. As well as quantisation to simulate real-world performance into 8-bit integer, to reduce model size and speed up inference on the smart glasses.

A selected pair of smart glasses capable of connecting to a external processing device for the models will be used. They will capture footage and send it to the models for processing and testing.

5. Comparative analysis

A final comparative analysis will then be done to summarise the results of the data collected. The comparison will be focused on the trade-off between accuracy and performance. However, will also include latency in real-time identification and mentions of the hardware feasibility and usability factors.

Data Collection Plan

Data Required	Date Expected	Support Required
Dataset of images at least 5 species of plants	15/11/25	Access to previous datasets of images
Dataset preparation involving the processing of the images	21/11/25	Access to image processing software
Selecting a convolutional neural network model	29/11/25	
Selecting a pre-trained learning model	29/11/25	
Selecting a lightweight model optimized for edge devices	29/11/25	
Training the convolutional neural network model	14/12/25	Access to programming software
Training the pre-trained learning model	30/11/25	Access to programming software
Training the lightweight model optimized for edge devices	17/11/25	Access to programming software
Model Evaluation	21/01/25	Access to programming software and evaluation logic
Model Optimisation	14/02/25	Access to programming software
Model Integration	01/03/25	Access to programming software and the 3D glasses

Comparative analysis	07/03/25	Help with analysis literature
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